Customer Shopping Behavior Analysis

1. Project Overview:

This project examines customer purchasing behavior using transactional data from 3,900 transactions across multiple product categories. Its objective is to identify insights related to spending trends, customer segmentation, product preferences, and subscription patterns to support informed strategic business decisions.

2. Dataset Summary:

- Rows: 3,900
- Columns: 18
- Key Features:
    - Customer demographics (Age, Gender, Location, Subscription Status)
    - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
    - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Data exploration and analysis carried out in Python:

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used df.info() to check structure and .describe() for summary statistics.

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

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 #   Column                  Non-Null Count  Dtype
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 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
```

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.

- **Feature Engineering:**
  - Created **age_group** column by binning customer ages.
  - Created **purchase_frequency_days** column from purchase data.

- **Data Consistency Check:** Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.

- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions):

Performed structured analysis in MySQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

gender	revenue
Male	157890
Female	75191

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79
33	67

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

item_purchased	average_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Handbag	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

shipping_type	average_amount
Standard	58.4602
Free Shipping	60.4104

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

subscription_status	total_customers	average_amount	total_amount
Yes	1053	59.49	62645
No	2847	59.87	170436

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

customer_segment	Number_of_customers
Loyal Customer	3116
Returning Customer	701
New Customer	83

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

category	item_purchased	total_orders	item_rank
Accessories	Jewelry	171	1
Accessories	Belt	161	2
Accessories	Sunglasses	161	3
Clothing	Pants	171	1
Clothing	Blouse	171	2
Clothing	Shirt	169	3
Footwear	Sandals	160	1
Footwear	Shoes	150	2
Footwear	Sneakers	145	3
Outerwear	Jacket	163	1
Outerwear	Coat	161	2

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

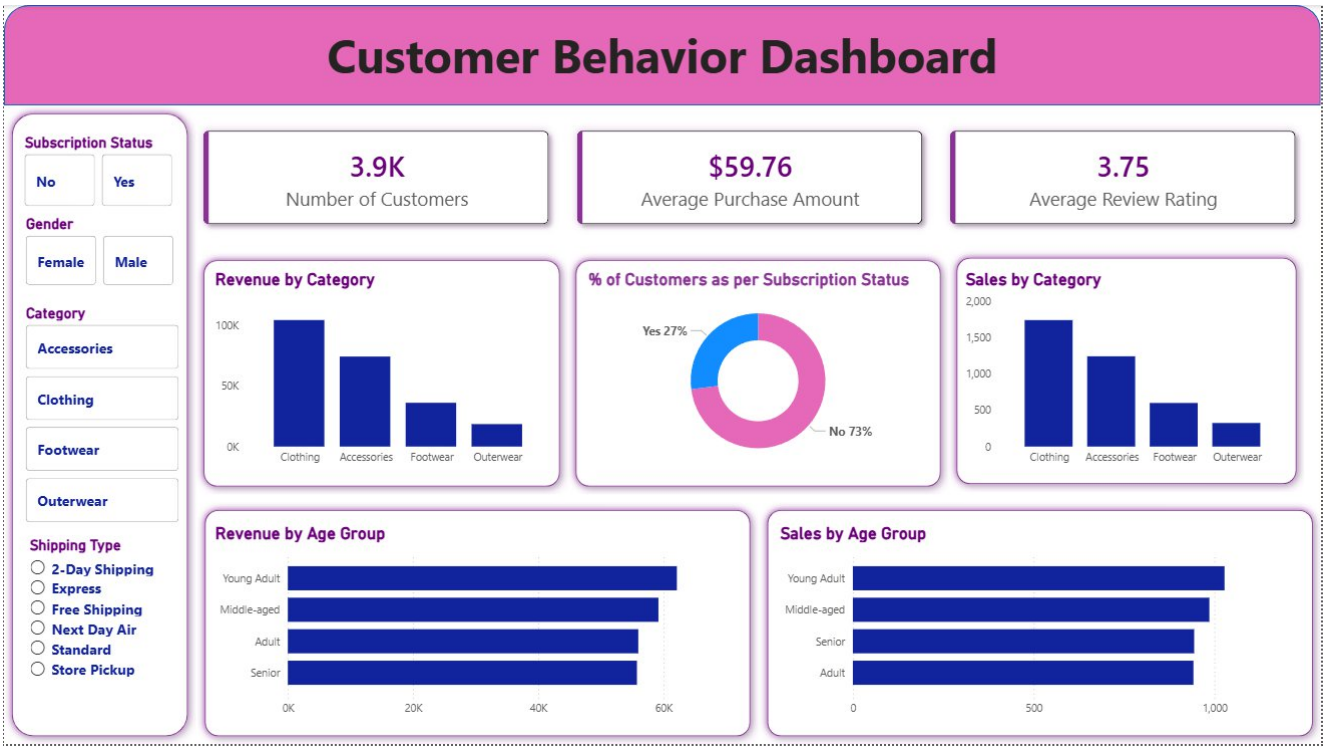
subscription_status	total_customers
Yes	958
No	2518

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

age_group	sales
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

5. Dashboard in Power BI:

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendation:

- **Increase Subscription Adoption** – Encourage customers to subscribe by promoting exclusive member benefits and added value features.
- **Enhance Customer Loyalty Programs** – Implement reward strategies to incentivize repeat purchases and transition customers into the “Loyal” segment.
- **Reassess Discount Strategy** – Optimize discount offerings to drive sales while maintaining healthy profit margins.

- **Strengthen Product Positioning** – Emphasize top-rated and best-selling products in marketing campaigns to maximize visibility and conversions.
- **Implement Targeted Marketing** – Direct promotional efforts toward high-revenue age groups and customers who prefer express shipping to improve campaign effectiveness.